

Abhinab Acharya

Machine Learning Researcher & AI Engineer

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Computer Science PhD researcher at RIT working on data-efficient deep learning, continual learning, and computer vision. First-author papers at ICML 2024 and NeurIPS 2026, a co-authored ICML 2026 paper, and industry experience building client-specific object detection systems and optimizing LLM inference.

Education

- Aug 2022 – Present **Rochester Institute of Technology**, PhD in Computer Science, expected May 2027
Relevant coursework: Statistical Machine Learning, Deep Learning, Data-Driven Knowledge Discovery, Non-Convex Optimization
- 2013 – 2017 **Tribhuvan University**, Bachelor's in Electrical Engineering, Lalitpur, Nepal
Relevant coursework: Artificial Neural Network, Probability and Statistics, Computer Programming, Microprocessors

Research and industry experience

- Jun 2026 – Present **Wasabi Technologies**, ML/AI Co-op, scheduled through Dec 2026
- **Object detection.** Build client-specific detection systems with DINO, TIPS, SAM, and OWL, achieving over 70% AP at 0.5 IoU while running at 3–5 FPS.
 - **LLM inference.** Optimize KV caching with NVIDIA Dynamo and cloud storage, reducing expensive storage requirements while achieving a 100% cache hit rate on the evaluated workload.
- Aug 2022 – Present **Rochester Institute of Technology**, Research Assistant
- NeurIPS 2026**
- **Generative AI.** Developed class-mixed diffusion augmentation using Stable Diffusion to generate prior-task data and counter shortcut learning, improving classification accuracy by 3 pp.
- ICML 2026**
- **Multimodal continual learning.** Co-developed a CLIP-based mixture-of-experts framework, improving classification accuracy by 4 pp.
- ICML 2024**
- **Data selection.** Developed BOSS, a size-aware one-shot subset selection method that balances sample diversity and difficulty, improving classification accuracy by up to 6 pp over state-of-the-art baselines.
- CVPR 2026**
- **Research mentoring.** Mentored video anomaly detection research using InternVL and CLIP on UCF-Crime and XD-Violence.
 - **Adversarial robustness.** Investigated layer-wise, vulnerability-aware fine-tuning that improved robustness by up to 17%, evaluating ViT-B/16 on CIFAR, ImageNet-R, and CUB under PGD attacks.
- 2019 – 2022 **North Star Developer's Village**, R&D Software Engineer
- Developed ML solutions for clinical-note text classification.
 - Deployed server-side functionality for remote learning platforms used by 50+ universities.

Technical skills

Methods	Computer vision, object detection, NLP, generative AI, diffusion models, vision-language models, large language models, LLM inference optimization, mixture of experts, fine-tuning
Models	CLIP, Stable Diffusion, ViT, ResNet, EfficientNet, InternVL, DINO, TIPS, SAM, OWL
Frameworks	PyTorch, scikit-learn, NumPy, pandas, SciPy, Matplotlib, LMCache, SGLang, vLLM, NVIDIA Dynamo, TensorRT-LLM
Infrastructure	KV cache optimization, cloud storage, Claude Code, Codex, Cursor
Languages	Python, C/C++, JavaScript, MATLAB, R, SQL
Web and data	Flask, React, MySQL, MongoDB

Publications

- NeurIPS 2026** **Class-Mixed Diffusion Augmentation for Shortcut-Breaking in Continual Learning**
Abhinab Acharya, Dayou Yu, Qi Yu, Xumin Liu
- ICML 2026** **Mixing Expertise with Confidence: A Mixture of Experts Framework for Robust Multi-Modal Continual Learning**
Md Abdullah Al Forhad, Yuansheng Zhu, **Abhinab Acharya**, Xumin Liu, Qi Yu, Weishi Shi
- ICML 2024** **Balancing Feature Similarity and Label Variability for Optimal Size-Aware One-shot Subset Selection**
Abhinab Acharya, Dayou Yu, Qi Yu, Xumin Liu

Teaching and leadership

- 2023 – 2024** **Rochester Institute of Technology**, Teaching Assistant
- Led lab sessions, graded coursework, and mentored students in Machine Learning, Computer Science, and Database Systems.
 - Built a data science learning platform (Flask, React, MongoDB) for Principles of Computing Immersion, supporting 100+ students; supervised its undergraduate contributors.
- 2022** **North Star Developer's Village**, Team Manager
- Managed a team of three developers researching and building educational robots.
- 2015 – 2017** **Tribhuvan University**, Robotics Team Leader
- Led the university team at ABU ROBOCON 2016 in Thailand.

Talks

- Nov 2024** **Data-Efficient Deep Learning**, GCCIS PhD Colloquium Series, RIT
- Jul 2024** **BOSS: Size-Aware One-shot Subset Selection**, poster presentation at ICML 2024

Peer review

- Conferences** 31 papers: NeurIPS 2026 (6) and 2025 (4); ICML 2026 (6); AAAI 2027 (5) and 2026 (4); ICLR 2026 (3); CVPR 2026 (1); ECCV 2026 (1); AISTATS 2026 (1)
- Journals** IEEE Transactions on Services Computing 2024 (1, including revision); TMLR 2026 (1)
- Other venues** IJCAI 2026; IEEE Big Data 2024; ACM AI Summit 2026
- Upcoming** Reviewer for ICLR 2027 and AISTATS 2027

Honors

- 2026** **Silver Reviewer Award**, ICML 2026; recognized among top reviewers based on area-chair ratings
- 2016** **Best Engineering and Panasonic Awards**, ABU ROBOCON 2016; team prize equivalent to US\$500